

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (17 marks – approximately 31 minutes)

Many companies and organisations have allowed their employees to work from home ("WFH") during the COVID-19 pandemic. These employees might use different remote communication and network connection technologies to complete tasks assigned by their bosses. Issues of data protection and systems security have been more and more critical and of concern in a WFH environment.

A report has mentioned that lax password protection and social engineering attacks are some of the new risks of WFH.

- (a) Define the term "social engineering".
(2 marks)
- (b) Demonstrate how attackers conduct social engineering attacks with any THREE examples.
(6 marks)
- (c) Analyse why the following measures could enhance the strength of password authentication:
 - (i) Requiring users to change their passwords every three months;
(3 marks)
 - (ii) Locking a user's account automatically if the user has provided an incorrect password three times consecutively for that user account in the login portal; and
(3 marks)
 - (iii) A new requirement to extend the length of a login password from 6 characters to 10 characters.
(3 marks)

Question 2 (19 marks – approximately 34 minutes)

- (a) In the retail industry, managers and proprietors think about how their company can perform better than its competitors in order to obtain competitive advantages.

Required:

State and describe any **THREE** areas of competitive advantage which a company can develop, achieve or maintain.

(6 mark)

- (b) Categorise the **THREE** types of reports which can be generated by a management information system ("MIS").

(3 marks)

- (c) Based on the answers in part (b), demonstrate each type of report with an example.

(6 marks)

- (d) Contrast the **TWO** differences between predictive data analytics and prescriptive data analytics.

(4 marks)

Question 3 (20 marks – approximately 36 minutes)

- (a) Summarise the primary and secondary activities under Porter's value chain model.

(3 marks)

- (b) Outline any **TWO** primary duties or responsibilities of a production department in a manufacturing company and any **TWO** types of its information needs.

(4 marks)

- (c) Classify any **THREE** basic functions of a computer network with brief descriptions.

(6 marks)

- (d) List and describe the **THREE** common cloud deployment models.

(3 marks)

- (e) List and describe any **TWO** benefits of using cloud storage services.

(4 marks)

Question 4 (11 marks – approximately 20 minutes)

- (a) The waterfall model is a popular and traditional system development life cycle approach that was used for software development. This model required significant system analysis and system design before starting system implementation.

However, programmers or system analysts now may consider adopting other software development models for their software development projects.

Required:

Apart from the waterfall model, state and describe any TWO other software development models.

(4 marks)

- (b) A common approach to evaluate suitability among different packaged accounting software is to identify and compare the features among the software comprehensively. To speed up this evaluation process, we can categorise the features into three types: essential features, value adding features and non-essential features.

Required:

Demonstrate any ONE essential feature and any ONE value adding feature of a packaged accounting software.

(4 marks)

- (c) List any THREE common sources of system change requests.

(3 marks)

Question 5 (13 marks – approximately 23 minutes)

- (a) State any **FOUR** reasons why Hong Kong is perfectly suited and situated for e-commerce imports and exports.

(4 marks)

- (b) Imagine that you are a general manager of a new online home appliance store. In a monthly meeting, you suggested selling some fancy and innovative home appliance products from Japan and South Korea in order to compete with other well-established retailers in the market. However, one of the directors worried that some customers may be displeased with these fancy and innovative products after receipt and that they may then ask to return the product during the guaranteed seven-day return period.

Required:

Justify the importance of the following areas that an e-commerce retailer should consider to reduce the possibility of receiving non-defect products from displeased customers:

- (i) Managing customer expectations; and

(3 marks)

- (ii) Training customers for better product knowledge.

(3 marks)

- (c) Knowledge management systems ("KMS") are usually used in customer service or support activities. A customer service representative can get reliable answer(s) or solution(s) quickly by simply typing a customer's question in the full-text search function of an internal KMS.

Required:

List the sequence of the SIX major steps under the knowledge management system cycle.

(3 marks)

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